Abdallah Essam

Mechatronics & Robotics Engineer

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PROFESSIONAL SUMMARY

Mechatronics Engineer with 5+ years of hands-on experience in robotics, autonomous systems, embedded computing, and AI-based perception. Experienced in developing robotic applications using **ROS 2**, **Python**, **Modern C++**, **MATLAB/Simulink**, **RViz**, and **Gazebo**, with hands-on work across articulated robots, autonomous ground vehicles, and UAV platforms. Skilled in computer vision, object detection using **YOLO/ONNX**, sensor integration, embedded systems, and real-time control, including deployment on **NVIDIA Jetson** platforms. Proven experience integrating robotic hardware, developing autonomous navigation and control applications, and supporting robotics and UAV teams. Holds a Bachelor's degree in **Automotive Mechatronics Engineering**, with an **A+ graduation project** focused on distributed embedded communication systems. Interested in robot learning, embodied AI, computer vision, and autonomous robotic systems, with a strong motivation to contribute to research-driven robotics development and translate advanced concepts into real-world robotic applications.

EXPERIENCE

Robotics & AI Specialist (Full-time) | Atlab, Dubai *Jul 2025 – Present*

- Develop and integrate robotic systems using **ROS 2**, **RViz**, and **Gazebo** across articulated and autonomous platforms, including the **Niryo Ned2** robotic arm and **Quanser QCar2** autonomous vehicle.
- Develop autonomous robotics applications using **Python**, **C++**, and **MATLAB/Simulink**, implementing path planning, control, and perception algorithms for deployment on **the NVIDIA Jetson AGX Orin** platform.
- Develop **computer vision** solutions for robotic **perception** using **YOLO** and **ONNX** models, enabling real-time object detection and perception-driven autonomous navigation.
- Configure and integrate **Unitree G1 humanoid** and **Unitree Go2 quadruped** robots, working with ROS 2 packages, **SDK** integration, **URDF** models, and simulation environments for robotic development and testing.
- Integrate and evaluate **UAV** platforms, including **flight control** systems, onboard sensors, and autonomous programming workflows for robotics applications.
- **Design** and deliver technical **training** in robotics, AI, autonomous systems, and embedded engineering for educators and engineering professionals.
- **Design** and prototype custom mechanical components, sensor enclosures, and robotic fixtures using **CAD** and **3D printing**, supporting rapid hardware iteration and robot integration.

Technical Mentor (Project-based) | UMEX Drone Challenge, Abu Dhabi *Nov 2025 – Jan 2026*

- Mentored university students and graduate engineers' participants in the design and implementation of autonomous **UAV** systems for competitive missions.
- Delivered technical guidance on UAV flight dynamics, autonomous navigation, and maze-solving algorithms.
- Supported AI-based perception using **ONNX** object detection models for real-time mission execution.
- Assisted teams in debugging flight control, computer vision, and mission-planning workflows under real competition constraints.

STEM & Robotics Instructor (Part-time) | Rsanh Educational Consultant, Cairo *Dec 2024 – Jul 2025*

- Teach robotics and STEM subjects to school students, focusing on practical applications and skill-building.
- Develop and design a robotics curriculum and hands-on activities for grades 9 and 10 in partner schools.
- Collaborate with school teams to ensure curriculum aligns with students' educational needs and standards.

Technical Project Manager (Project-based) | Global Robotics Challenge GRC, Cairo *Oct 2024 – Jun 2025*

- Led the technical development of the GRC competition, ensuring all challenges met high engineering and educational standards.
- Managed and coordinated over 1,000 participants across 150 teams, ensuring smooth execution of all technical activities and fair competition processes.
- Developed new competition challenges focused on real-world applications including robot manipulation, pick-and-place, and autonomous navigation to build participants' technical skills.
- Created detailed rulebooks for each challenge, ensuring clarity and fairness across all competition categories.
- Managed the technical requirements for the competition venue, ensuring full compliance with mechanical design specifications.
- Produced all competition arena and component designs in OnShape and SolidWorks, maintaining versioned CAD libraries accessible to the full technical team.

Robotics Instructor (Full-time) | Discovery AST, Cairo *Aug 2023 – Jul 2025*

- Delivered hands-on instruction in Arduino programming, Raspberry Pi, PCB design, IoT implementation using ESP32 and Arduino Cloud IDE.
- Trained students in **SolidWorks** for robotic component design and 3D modeling.
- Coached national and international robotics competition teams.
- Used **OnShape** and **SolidWorks** for designing and iterating on robot components, teaching CAD-to-fabrication workflows to students across multiple project cycles.

STEM & Robotics Instructor (Part-time) | Engineeius, Cairo *Oct 2022 – Jul 2023*

- Instructed students in Arduino, robot design, motors, and communication protocols.
- Guided students in developing entrepreneurial skills, including business planning and project presentation.
- Fostered a collaborative and creative learning environment, emphasizing teamwork and problem-solving abilities.

Founder & Mechanical Engineer | Pegasi Robotics Team, Cairo *Dec 2021 – May 2022*

- Designed and optimized robot motors for torque and performance.
- Led the mechanical design, ensuring robust and efficient structural integrity.
- Selected materials and designs for weapon systems to maximize efficiency and durability.
- Managed the manufacturing and assembly process, ensuring compliance with competition regulations.

EDUCATION

Bachelor of Automotive Mechatronics Engineering *GPA: 3.11 / 4.0*

Helwan University, Mataria Branch *2019 – 2024*

Graduation Project: Adaptive AUTOSAR Communication Stack for Infotainment Systems (Grade: A+)

- Designed a scalable infotainment application on Adaptive AUTOSAR architecture with Ethernet-based SOME/IP inter-process communication on Raspberry Pi.
- Integrated MQTT protocol for lightweight publish/subscribe messaging across distributed embedded components.
- Simulated real-time automotive network behavior across distributed software components on embedded Linux.

TECHNICAL SKILLS

Robotics & Control:	ROS2, Gazebo, RViz, PID control, real-time closed-loop systems, UAV flight control
CAD & Design:	SolidWorks, Fusion 360, OnShape (familiar), 3D printing & prototyping
Hardware:	NVIDIA Jetson AGX Orin, STM32, ESP32, AVR, Arduino, Raspberry Pi, PCB design, HAL Layer interfacing, motor control (DC, stepper, servo)
Programming:	Modern C++ (11/14/17), Embedded C, Python, Bash scripting
Communication:	Camera integration, IMU, UART, I^2C, SPI, CAN, LIN, MQTT, Ethernet, SOME/IP
Software & Tools:	MATLAB, Simulink, STM32CubeIDE, VS Code, Linux (Ubuntu), Git, RTOS, OOP, SOLID principles, Design Patterns
AI & Computer Vision:	OpenCV, YOLO, ONNX Runtime, object detection, image processing, perception systems,

CERTIFICATIONS

- STEM.org Certified Master Trainer
- ROS 2 Foundations – Packt
- AUTOSAR MasterClass – SPRINTS
- Embedded C++ Diploma – Robotics Corner
- Robotics Workshop – BETA Academy
- Entrepreneurship Program for Youth –TIEC – InnovEgypt

AWARDS & RECOGNITION

- Finalist – ISEF Competition 2017
- 4th Place – Metal Monsters Competition 2022
- PR Member at YLY
- Events Planner at Astronomy Club-ASU
- 2nd Place – NASA Space Apps Cairo 2019
- Logistics Director, TEDx
- Entrepreneurship Trainee at SCCI

LANGUAGES

Arabic: Native | **English:** C1 (Fluent)